MarketIQ

Marketing Intelligence Platform

Document Type	Version	Status
Product Requirements Document	v2.0 — Full Release	**Approved for Development**
Date	**Cloud**	**AI Engine**
April 2025	Google Cloud Platform (GCP)	**Claude API (Anthropic)**

Mission: Enable marketing agencies and enterprises to make faster, smarter decisions through unified data intelligence and AI-powered insights.

Build target: 20 weeks to first revenue (MVP) · 26 weeks to full v1 platform

Table of Contents

1. Executive Summary

MarketIQ is a white-label, multi-tenant B2B SaaS marketing intelligence platform built for marketing agencies and enterprise organisations. It unifies data from every major marketing channel into a single normalised model, then surfaces performance insights and recommendations through an AI-powered dynamic dashboard builder.

The platform is fully invisible to end users — agencies and enterprises white-label it under their own brand. Customers interact with their brand, not MarketIQ. The underlying intelligence engine is powered by Claude (Anthropic), running on Google Cloud Platform with Snowflake as the analytics warehouse.

1.1 Key Differentiators

Differentiator	Description
AI Dynamic Dashboards	Natural language to fully rendered dashboards in seconds, powered by Claude API
True White-Labeling	Complete brand replacement — agencies resell the platform as their own product
Dual Tenant Architecture	Agency + Enterprise topologies supported simultaneously on one shared platform
Universal Data Model	Single normalised marketing schema across all channels — one query language for everything
Configuration-First Design	5-day onboarding target — zero custom code, config-driven tenant provisioning
Cost-Optimised Infrastructure	GCP-native stack — 35–40% lower infra cost vs Azure equivalents at scale

1.2 Platform Summary

Dimension	Agency Mode	Enterprise Mode
Tenant is	The Agency	The Company
Sub-tenants are	Clients	Brands / Regions / Business Units
White-label to	Agency's clients	Internal teams
Dashboard scope	Per client	Per brand/region/BU
Key permission	Client data isolation	RBAC by org hierarchy
Top use case	Reporting to clients under own brand	CMO to BU performance visibility

2. Target Market & Customer Profiles

2.1 Primary Segments

Segment	Target Customer	Priority	Launch Phase
Digital Agencies	Performance/media agencies managing multiple client accounts	P0 — Critical	MVP
Enterprise Marketing	In-house marketing at mid-to-large companies	P0 — Critical	MVP
Marketing Consultancies	Boutique firms needing white-labeled reporting tools	P1 — High	v1
Media Buying Firms	Specialist paid media agencies with heavy channel mix	P1 — High	v1

2.2 Buyer Personas

Agency Buyer

Attribute	Detail
Typical Title	Agency Owner · Head of Performance Marketing · VP Digital
Primary Pain	Manual client reporting consuming 20%+ of team time
Secondary Pain	Fragmented channel data — no single source of truth per client
Buying Trigger	Losing clients to agencies with better insight delivery
White-label Value	Resell platform under own brand — increases perceived agency value
Success Metric	Client retention rate · Hours saved on reporting per month

Enterprise Buyer

Attribute	Detail
Typical Title	CMO · VP Marketing · Marketing Operations Director
Primary Pain	Siloed channel data — Google Ads, Meta, Salesforce never reconciled
Secondary Pain	Board pressure for marketing ROI accountability with no unified view
Buying Trigger	Budget review cycle or new CMO seeking better visibility
White-label Value	Internal branding improves tool adoption across regional teams
Success Metric	ROAS improvement · Budget optimisation · Speed of insight-to-action

3. Core Features

3.1 White-Labeling

Every tenant fully replaces the platform's visual identity. The MarketIQ brand is entirely invisible to end users. Agencies can resell the platform as their own product; enterprises apply their internal branding.

Visual Identity & Branding

- Custom logo — header, favicon, and email templates
- Brand colour system — primary, secondary, accent applied via CSS variables across entire UI
- Custom subdomain routing: analytics.theiragency.com with wildcard SSL
- Branded login page, transactional email sender name, and notification templates
- Dashboard widget themes inherit tenant brand colours automatically

Domain & Access

- Custom subdomain per tenant via Google Cloud CDN wildcard DNS
- SSO integration — Azure AD, Okta, Google Workspace per tenant
- Enterprise tenants can bring their own identity provider
- White-label mode flag per tenant — when enabled, all MarketIQ branding is suppressed

Tenant Configuration Object

```
tenant_id, logo_url, favicon_url,
brand_colors { primary, secondary, accent, background },
custom_domain, sso_config { provider, client_id, tenant_id },
white_label_mode: true | false,
email_sender_name, default_currency, default_timezone, locale
```

3.2 Multi-Tenancy

The platform implements hierarchical multi-tenant architecture. Both Agency and Enterprise topologies run on shared infrastructure with complete data isolation enforced at the database layer.

Data Isolation

- Snowflake workspace-per-tenant — data never resides in a shared schema
- Row-level security injected at query time — every query carries tenant_id + scope filter
- Sub-tenant (client/brand) data is siloed within its parent tenant
- Cross-tenant queries are architecturally impossible — not just access-controlled

Automated Tenant Provisioning

Step	Action
1	Create Snowflake workspace and seed Bronze / Silver / Gold schemas
2	Create tenant config object with branding defaults and locale settings
3	Seed RBAC structure — Admin role assigned to founding user automatically
4	Configure sub-tenant topology (clients for Agency; brands/regions for Enterprise)
5	Assign custom subdomain and provision wildcard SSL certificate
6	Configure SSO if enterprise plan is selected
7	Tenant is ready for connector authentication

3.3 Internationalisation (i18n)

English-first at launch. The i18n framework is fully in place from day one — adding a new language requires translation files only, no code changes.

Layer	Scope in v1	Implementation
Layer 1 — UI/UX	All menus, labels, tooltips, date formats, number formats, currency display	react-i18next with JSON locale files per language
Layer 2 — AI (partial)	Dashboard headers and KPI label names only	Static KPI dictionary + Azure Translator API for dynamic strings generated by AI
Layer 3 — Data Intelligence	Out of scope — no ad copy sentiment or local benchmarks	Not applicable in v1

3.4 Connector Framework

All connectors implement a standard interface — authenticate(), extract(), normalize(), schedule(), validate(). Airbyte handles commodity API extraction; a proprietary normalization layer maps all raw data to the Universal Marketing Schema.

Tier 1 — Launch Connectors (MVP)

Connector	Data Ingested	Criticality	MVP Timeline
Google Ads	Spend, impressions, clicks, conversions, ROAS, Quality Score	Critical	Week 7
Meta Ads	Spend, reach, frequency, CPM, video views, audience insights	Critical	Week 8
Google Analytics 4	Sessions, goals, attribution, user behaviour, conversion paths	High	Week 9
HubSpot	Leads, pipeline stages, campaign performance, deal attribution	High	Week 9
Salesforce	Revenue attribution, closed-won, pipeline, campaign influence	High	Week 10

Tier 2 — First 6 Months Post-Launch

Connector	Target Segment	Key Data	Target Month
LinkedIn Ads	B2B enterprise	Spend, leads, company targeting, InMail	Month 3
TikTok Ads	Consumer brands, agencies	Spend, video views, conversions	Month 3
Klaviyo	E-commerce, email marketing	Opens, clicks, revenue attributed	Month 4
Shopify	E-commerce brands	Orders, revenue, product performance	Month 5
Google Search Console	All segments — SEO	Impressions, clicks, position, CTR	Month 6

Connector Architecture

Data flow: External API → Airbyte (raw extraction) → Normalisation Layer → Universal Schema → Snowflake Bronze

Airbyte handles: API pagination · Rate limiting · OAuth token refresh · Retry logic

Normalisation layer handles: Field mapping · Currency conversion · Deduplication · Schema enforcement

The normalisation layer is proprietary IP — this is where MarketIQ's data understanding lives.

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3.5 Universal Marketing Data Model

Every data source is normalised to a single schema at ingestion time. This enables cross-channel comparison, unified attribution, and AI querying without per-channel logic in any dashboard or query.

Core Entities

Entity	Key Fields	Layer	Notes
CAMPAIGN	campaign_id, tenant_id, channel, objective, status, budget, currency	Silver	Normalised across all channels
AD_GROUP	ad_group_id, campaign_id, targeting_summary, bid_strategy	Silver	Maps to Ad Set (Meta), Ad Group (Google)
CREATIVE	creative_id, format, headline, body, destination_url	Silver	All creative types unified
DAILY_PERFORMANCE	date, tenant_id, campaign_id, spend, impressions, clicks, conversions	Gold	Primary fact table — all queries run here
CALCULATED_METRICS	CTR, CPC, CPM, CPA, ROAS, CVR	Gold	Derived — never stored raw in source

Gold Layer Tables

Gold Table	Purpose & Contents
GOLD_CAMPAIGN_DAILY	Pre-aggregated performance per campaign per day · Normalised to tenant base currency · All calculated metrics included
GOLD_CHANNEL_SUMMARY	Cross-channel roll-up · Budget vs actual vs pacing · All channels in one row per tenant per day
GOLD_TREND_BASELINE	Rolling 4-week, 8-week, 12-week averages per KPI per channel · Powers all anomaly detection

3.6 AI Dynamic Dashboard Engine

The dashboard engine is MarketIQ's primary differentiator. Users create fully rendered, data-populated dashboards through natural language prompts, manual widget building, or a hybrid of both on the same shared canvas.

Creation Modes

Mode	Description	Best For
Prompt to Dashboard	Type a natural language description. AI parses intent, generates a full dashboard spec, runs all queries in parallel, renders complete dashboard.	Power users, executives, fast insights
Widget by Widget	Select a widget type from the panel, configure via guided form with AI suggestions, preview with live data, add to canvas.	Analysts, detailed builders, non-technical users
Template Start	Choose a pre-built or saved template. AI auto-populates with tenant's live data. All widgets remain individually editable.	New users, consistent reporting formats

AI Pipeline — Prompt to Dashboard

Stage	What Happens
1. Intent Parsing	Claude extracts: metrics requested, dimensions, time range, filters, display preferences, special instructions (e.g. 'highlight underperforming')
2. Spec Generation	Claude generates a structured JSON dashboard spec — widget types, positions, configs, conditional formatting rules
3. Query Generation	Each widget spec maps to a parameterised SQL query against the Gold layer — tenant_id and scope injected automatically
4. Parallel Execution	All widget queries run simultaneously against Snowflake — no sequential waterfall
5. Render	Spec + query results → React rendering engine → live dashboard with charts, tables, KPI cards
6. Anomaly Scan	Results compared against GOLD_TREND_BASELINE — anomalies surfaced as inline overlays

Widget Types

Widget	Best Use Case	Key Configuration
KPI Card	Single metric headline with period comparison	Metric, comparison period, trend arrow, threshold colouring
Bar Chart	Comparing a metric across a dimension	Metric, dimension, sort order, value labels on bars
Line Chart	Trends over time — single or multi-series	Metric, time grain (day/week/month), group-by channel

Pie / Donut	Share of total — e.g. spend by channel	Metric, dimension, max slices, centre label
Data Table	Detailed sortable data with conditional formatting	Columns, default sort, threshold-based row colour rules
Funnel Chart	Conversion stage analysis with drop-off rates	Stage labels, values, show drop-off percentage
Heatmap	Two-dimension intensity — e.g. day-of-week vs channel	X axis, Y axis, intensity metric, colour scale
Scatter Plot	Correlation between two metrics	X metric, Y metric, optional bubble size metric
Alert Card	Anomaly or threshold breach flag	Metric, threshold rule, baseline comparison period
Text / Annotation	Section headers, notes, context blocks	Free text, markdown formatting

Dashboard Save & Reuse System

State	Description & Access
Draft	Being built — not saved · Only creator sees it · No auto-refresh
Saved	Saved to personal library · Only owner sees it · Live data refresh on open
Shared	Visible to assigned team or full org scope · Owner controls edit permissions
Published	Pinned to tenant or client homepage · All users in scope see it automatically
Scheduled	Auto-refreshes on cadence · Emails PDF snapshot to recipient list
Template	Available as starting point in template library · No live data · Structure only

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3.7 Recommendation & Anomaly Engine

Every dashboard render automatically scans results against the GOLD_TREND_BASELINE and surfaces inline recommendations. Rules-based at launch; ML-enhanced in v2.

Signal Type	Examples & Logic
Red — Critical Anomaly	Metric is >50% above/below 4-week rolling average · Budget >90% spent mid-month · Campaign inactive for 14+ days
Amber — Warning	Metric trending in wrong direction for 3+ consecutive days · Budget pacing will overspend if trend continues · CTR declining week-over-week
Green — Positive Signal	Metric above 4-week average by >20% · ROAS above tenant-defined target · Best performing channel flag

4. Permissions & RBAC

MarketIQ implements hierarchical multi-tenant RBAC. Access is defined across two dimensions: Role (what actions you can perform) and Scope (which data you can see). Every query execution automatically enforces both dimensions.

4.1 Role Hierarchy

Role	Level	Agency Context	Enterprise Context
Super Admin	Platform	MarketIQ team only — full platform, all tenants	MarketIQ team only
Tenant Admin	Tenant	Agency admin — all clients, billing, SSO, branding	Enterprise admin — full company, all brands
Manager	Sub-tenant	Senior analyst — all clients, connect data sources	Brand/Region manager — own scope only
Analyst	Sub-tenant	Build + edit dashboards for all clients	Build + edit dashboards within assigned scope
Viewer	Sub-tenant	View dashboards only — all clients, read-only	View dashboards only within assigned scope
Client User	Sub-tenant	Client sees own data only via white-labeled platform	Not applicable in Enterprise mode

4.2 Permission Matrix

Action	Admin	Analyst / Manager	Viewer
View dashboards	All tenants/scopes	Within assigned scope	Within assigned scope
Build dashboards	Yes	Yes	No
Edit dashboards	Yes	Own dashboards only	No
Delete dashboards	Yes	Own dashboards only	No
Connect data sources	Yes	No	No
Invite users	Yes	No	No
Manage roles & scopes	Yes	No	No
View raw data	Yes	Yes	No
Export data (CSV/PDF)	Yes	Yes	Yes
Manage billing	Yes	No	No
Configure white-label	Yes	No	No

Schedule reports	Yes	Yes	No
Impersonate users	Super Admin only	No	No

4.3 Data Scoping & Row-Level Security

Roles control actions. Scopes control what data is visible. The scope filter is injected at the query layer — it is invisible to users and cannot be bypassed.

```
User Object:    { user_id, tenant_id, role: 'analyst', scope: { type: 'client', ids: ['client_A'] } }

Every executed query automatically becomes:
  SELECT ... FROM gold_campaign_daily
  WHERE tenant_id = :tenant_id            ← tenant isolation
  AND   client_id IN ('client_A')         ← scope filter

Users cannot see data outside their scope even if they construct a custom query.
```


5. Onboarding Flow

Target: sign-up to first live AI dashboard in 5 days or fewer. Every onboarding step is designed around this goal. No manual intervention is required from the MarketIQ team for standard onboarding.

5.1 Five-Step Journey

Step	Name	Target Day	Goal & Exit Criteria
Step 1	Account & Profile Setup	Day 1	Account created · Tenant type selected · Org structure defined · Branding defaults set
Step 2	Connect Data Sources	Day 1–2	At least one Tier 1 connector authenticated · Historical backfill configured and queued
Step 3	Platform Auto-Setup	Day 2–3	Bronze→Silver→Gold pipelines complete · GOLD_TREND_BASELINE seeded · AI pre-analysis done · User notified by email
Step 4	First Dashboard	Day 3–4	User enters platform · Sees personalised observations · Creates or selects first live dashboard
Step 5	Invite & Share	Day 4–5	At least one team member invited · First dashboard shared or scheduled

5.2 Automated Provisioning Pipeline

Triggered automatically on signup. Runs entirely without human intervention:

1. Snowflake workspace created and all schemas seeded (Bronze, Silver, Gold)
2. Tenant config object created with defaults — branding, locale, currency, timezone
3. RBAC structure seeded — Admin role assigned to founding user automatically
4. Sub-tenant topology configured — clients (Agency) or brands/regions (Enterprise)
5. Custom subdomain provisioned — wildcard SSL certificate assigned
6. SSO configured if Enterprise plan is selected
7. On first connector authentication: Airbyte pipeline spun up per connector
8. Historical backfill queued — configurable 3, 12, or 24 months of data
9. Bronze → Silver → Gold transformation jobs scheduled on completion
10. AI pre-analysis runs — personalised observations and starter dashboards generated
11. User notified via email: 'Your data is ready — here's what we found'

5.3 Onboarding Success Metrics

Metric	Target

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Sign-up to first connector connected	Less than 30 minutes
First connector to data ready	Less than 24 hours
Data ready to first dashboard saved	Less than 2 hours
Sign-up to first live dashboard	5 days maximum
Onboarding completion rate	Greater than 70%
Time to second dashboard created	Less than 7 days after first

6. Technical Architecture

6.1 Architecture Pattern

MarketIQ is built as a microservices platform with a serverless-first deployment model on GCP. All services are independently deployable, scale to zero when idle, and communicate via GraphQL Federation (synchronous, user-facing) or Cloud Pub/Sub (asynchronous, background).

> Synchronous path: Frontend → Apollo Federation Gateway → Subgraph services → Response
> Asynchronous path: Service → Cloud Pub/Sub → Cloud Function worker → Side effect
>
> Rule: Users never wait for background work. All heavy processing is event-driven and asynchronous.

6.2 Service Inventory

Core API Services — Cloud Run (scale to zero)

Service	Responsibility
Apollo Federation Gateway	Single GraphQL entry point — routes to all subgraph services via schema stitching
Auth Service	Firebase Auth wrapper · Token validation · Session management · SSO integration
Tenant Service	Tenant provisioning · Config management · Sub-tenant CRUD · Branding config
User Service	User management · Role assignment · Scope management · Invitation flow
Dashboard Service	Dashboard spec storage · Save / share / publish / template system
Widget Service	Widget configuration · Type registry · Preview data generation
Query Service	Snowflake query execution · Gold layer access · Redis query caching
AI Service	Claude API orchestration · Prompt → spec generation · Intent parsing · Spec → SQL
Connector Service	Connector config management · OAuth flows · Sync scheduling · Status tracking
Recommendation Service	Anomaly detection · Baseline comparison · Insight generation · Alert rules

Event-Driven Workers — Cloud Functions

Worker	Trigger & Responsibility
Sync Trigger	Scheduled (configurable) — fires Airbyte connector data pulls per tenant per connector
Export Worker	Event-triggered — PDF generation for scheduled dashboard snapshot emails

Notification Worker	Event-triggered — transactional email, anomaly alerts, onboarding milestone nudges
Provisioning Worker	Event-triggered on signup — automated tenant setup pipeline execution
Baseline Computer	Scheduled nightly — recomputes GOLD_TREND_BASELINE rolling averages for all tenants

7. Technology Stack

7.1 Frontend

Technology	Package	Purpose	Rationale
React + TypeScript	react@latest	UI framework	TypeScript throughout — matches backend types
Vite	vite@latest	Build tool	Fast HMR, optimised production builds
Apollo Client	@apollo/client	GraphQL client	Native React hooks, caching, optimistic UI
shadcn/ui	shadcn/ui	Component library	Unstyled — critical for white-label theming
Recharts	recharts	Standard charts	React-native, composable, lightweight
Apache ECharts	echarts-for-react	Complex charts	Heatmap, funnel, scatter — not in Recharts
react-grid-layout	react-grid-layout	Dashboard canvas	Drag, drop, resize — saves layout as JSON spec
Zustand	zustand	Global UI state	Lightweight — tenant context, theme, sidebar
TanStack Query	@tanstack/react-query	Server state	Per-widget data fetching, caching, refresh
react-i18next	react-i18next	i18n	English-first, locale-ready, lazy loads files
GraphQL Code Generator	@graphql-codegen/cli	Type generation	Auto-generates TypeScript types from schema

7.2 Backend

Technology	Package	Purpose	Rationale
Node.js + TypeScript	node@20	Runtime	All services — unified language with frontend
Apollo Server	@apollo/server	GraphQL	One subgraph server per microservice
Apollo Federation	@apollo/subgraph	Schema stitching	Gateway merges all subgraphs seamlessly
Pothos	@pothos/core	Schema builder	Code-first, TypeScript-native — no SDL drift

Prisma	prisma	ORM	Type-safe, per-service schemas, auto-migrations
Zod	zod	Validation	Input schemas, shared with frontend via packages/
BullMQ	bullmq	Job queues	Redis-backed, reliable retry, rate limiting
Anthropic SDK	@anthropic-ai/sdk	Claude API	Dashboard spec generation, intent parsing

7.3 Data Layer

Technology	Role	Analytics Layer	Notes
Snowflake (on GCP)	Analytics warehouse	Bronze + Silver + Gold	Consumption pricing, multi-tenant workspaces
Google Cloud Storage	Data lake	Bronze (raw)	Immutable raw data — replayed if pipelines fail
dbt	Transformations	Silver + Gold	Version-controlled SQL — Bronze → Silver → Gold
Apache Airflow	Orchestration	Pipeline scheduling	Cloud Composer (managed) for production
Airbyte	Connector extraction	Bronze ingestion	Wrapped by proprietary normalization layer
Cloud SQL PostgreSQL	Operational database	Application state	Tenant configs, users, dashboard specs, audit
Memorystore Redis	Cache + queues	App layer	Query result cache, BullMQ backend, rate limits

7.4 Infrastructure — GCP

GCP Service	Purpose
Cloud Run	All microservices — auto-scales to zero when idle
Cloud Functions	Event-driven workers — sync triggers, exports, notifications
Cloud Pub/Sub	Async messaging between services — durable, at-least-once delivery
Firebase Auth	Multi-tenant authentication · SSO integration per tenant
Cloud CDN + Cloud Load Balancing	Global CDN · White-label subdomain routing · SSL termination
Cloud Armor	WAF + DDoS protection on all public endpoints
Secret Manager	Connector OAuth tokens + API keys — encrypted at rest

Artifact Registry	Private Docker image storage — one repo per service
Cloud Monitoring + Cloud Logging	Infrastructure metrics, alert policies, centralised log aggregation
Sentry	Frontend + backend error tracking with full stack traces

7.5 DevOps & Tooling

Tool	Purpose
Turborepo	Monorepo management — shared packages, incremental build caching
GitHub + GitHub Actions	Source control + CI — run tests, build Docker images, push to Artifact Registry
Terraform	GCP infrastructure as code — tenant provisioning scripts included
Docker	Containerisation for all services — consistent dev and prod environments
Puppeteer	Headless Chrome PDF export — runs as Cloud Function, triggered by job queue
Resend	Transactional email — developer-friendly API, React Email templates

8. Repository Structure

Monorepo managed by Turborepo. All apps, services, workers, shared packages, data pipelines, and infrastructure live in a single repository with shared TypeScript configs and ESLint rules.

```
/ (root — Turborepo monorepo)

apps/
  web/              ← React + Vite frontend application
  gateway/          ← Apollo Federation Router (GraphQL gateway)
  services/
    auth/           ← Auth microservice + Firebase wrapper
    tenant/         ← Tenant provisioning + config management
    user/           ← User management + RBAC
    dashboard/      ← Dashboard specs + save / share / template
    widget/         ← Widget config registry + preview
    query/          ← Snowflake query engine + caching
    ai/             ← Claude API orchestration + spec generation
    connector/      ← Connector management + OAuth flows
    recommendation/ ← Anomaly detection + insight engine

workers/
  sync-trigger/     ← Cloud Function: fires connector pulls
  export/           ← Cloud Function: PDF generation
  notification/     ← Cloud Function: email + alerts
  provisioning/     ← Cloud Function: tenant setup pipeline
  baseline/         ← Cloud Function: Gold layer aggregations

packages/
  db/               ← Prisma schema + migration files
  snowflake/        ← Snowflake client + typed query builders
  types/            ← Shared TypeScript type definitions
  ui/               ← shadcn/ui component library (white-label aware)
  config/           ← Shared ESLint + TypeScript configs

data/
  dbt/              ← dbt models (Bronze → Silver → Gold)
  airflow/          ← Airflow DAGs (pipeline orchestration)
  airbyte/          ← Connector configuration files

infra/
  terraform/        ← GCP infrastructure as code (all environments)
```

9. Development Timeline

The timeline below assumes Claude Code as the primary development accelerator (reducing boilerplate and scaffolding time by approximately 35–40%), and a solo founder or 1–2 developer team starting from zero. Two milestones structure the roadmap: MVP at Week 20 (first paying customers) and Full v1 at Week 26.

> Key milestones:
> Week 2 — Development environment fully operational
> Week 5 — Auth, tenant provisioning, and frontend shell complete
> Week 9 — Data layer live — first two connectors syncing to Gold layer
> Week 14 — Dashboard engine complete — manual widget builder fully functional
> Week 17 — AI layer live — prompt-to-dashboard end-to-end working
> Week 20 — MVP complete — first beta customers onboarded, first revenue target
> Week 26 — Full v1 complete — all connectors, white-label subdomains, enterprise RBAC

P0 Environment & Infrastructure Setup — Weeks 1 – 2

This phase establishes the development foundation. Nothing can be built without it. Time invested here pays back every day of the project.

Week	Tasks & Deliverables
Week 1	GCP project creation + IAM setup · GitHub organisation + monorepo scaffold (Turborepo) · Turborepo workspace configuration (apps/, packages/, workers/) · Docker + local dev environment · CI pipeline skeleton (GitHub Actions — lint + test on PR)
Week 2	Terraform GCP modules (Cloud Run, Cloud SQL, Redis, Pub/Sub, Cloud Functions) · Snowflake account creation + workspace structure · Firebase Auth project setup · Cloud CDN + load balancer config · Artifact Registry setup · Staging environment first deploy

Phase 0 Exit Criteria

- Local developer environment working end-to-end
- Staging GCP environment deployed and accessible
- GitHub Actions running on every PR (lint, type-check, test)
- Turborepo build cache working — shared packages compiling

P1 Foundation — Auth, Tenant, Frontend Shell — Weeks 3 – 5

The authentication, tenant topology, and UI shell must be solid before any feature work begins. White-label theming is built here — retrofitting it later is expensive.

Week	Tasks & Deliverables

Week 3	Auth service — Firebase Auth integration · JWT validation middleware · MSAL SSO hook for enterprise · Tenant service — provisioning API + Terraform automation · Prisma schema v1 (tenants, users, roles, scopes) · Cloud SQL PostgreSQL live
Week 4	User service — RBAC engine · Role assignment + scope management · Frontend shell — React + Vite + Turborepo · Apollo Client setup + GraphQL Code Generator · shadcn/ui baseline component library · CSS variable theming system (white-label ready from day one)
Week 5	Agency + Enterprise tenant profile setup flows (UI) · Sub-tenant creation (clients / brands / regions) · Sidebar navigation + layout · Tenant config loader — branding applied at auth time · Protected routes + role-based UI hiding · End-to-end auth flow tested

Phase 1 Exit Criteria

- A new Agency tenant can be provisioned in < 5 minutes via automated pipeline
- User can log in, see their tenant branding, and navigate the shell
- Role-based access control working — Admin sees all, Viewer is restricted
- White-label colours apply correctly from tenant config — verified in browser

P2 Data Layer — Snowflake, Pipelines, Connectors — Weeks 6 – 9

The data layer is the most technically complex phase. Snowflake, dbt, Airflow, and Airbyte are four distinct tools each with their own configuration and quirks. Plan for integration debugging time.

Week	Tasks & Deliverables
Week 6	Snowflake workspace structure — one database per tenant + shared utility database · dbt project setup — Bronze, Silver, Gold layers defined · Bronze schema DDL — raw ingestion tables for all Tier 1 sources · GCS data lake bucket structure per tenant · Airflow (Cloud Composer) setup + first test DAG
Week 7	Airbyte self-hosted setup (or Cloud) · Google Ads connector — Airbyte config + OAuth flow UI · Custom normalisation layer — Google Ads fields mapped to Universal Schema · Silver dbt models for Google Ads · GOLD_CAMPAIGN_DAILY initial model · First end-to-end sync tested
Week 8	Meta Ads connector — same pipeline as Week 7 · Universal schema gaps identified and filled · Silver deduplication logic · Currency normalisation (DIM_CURRENCY table) · GOLD_CHANNEL_SUMMARY model · Connector Service — GraphQL subgraph + sync status API
Week 9	GA4 connector · HubSpot connector · Salesforce connector (partial — campaigns and pipeline) · GOLD_TREND_BASELINE model — rolling 4/8/12 week averages · Query Service — Snowflake query executor with Redis caching · Parameterised query builder — tenant_id + scope injection · Performance test: Gold layer query < 2 seconds p95

Phase 2 Exit Criteria

- Google Ads and Meta syncing live data into Gold layer for at least one tenant
- GOLD_CAMPAIGN_DAILY, GOLD_CHANNEL_SUMMARY, GOLD_TREND_BASELINE all populated
- Query Service executing parameterised Gold queries in < 2 seconds p95
- Connector OAuth flow working in-app — no external redirects

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P3 Dashboard Engine — Widgets, Canvas, Save/Share

Weeks 10 – 14

The dashboard engine is the product. This is the most substantial UI phase — react-grid-layout, charting libraries, live data, and tenant theming all intersect here. Build the manual widget builder before the AI layer.

Week	Tasks & Deliverables
Week 10	Dashboard Service — GraphQL subgraph · Dashboard spec JSON schema defined · Prisma schema for dashboards + widgets · react-grid-layout canvas scaffold — drag, drop, resize · Widget type registry — 10 widget types registered · KPI Card and Bar Chart widget components (Recharts)
Week 11	Line Chart widget (Recharts) · Pie / Donut widget · Data Table widget with conditional formatting · Widget config panel — right sidebar, step flow (pick type → configure → preview) · Live preview with real Snowflake data (via Query Service) · AI suggest chart type — lightweight Claude call in widget panel
Week 12	Funnel, Heatmap, Scatter, Alert Card, Text/Annotation widgets · Widget position + resize persistence (spec JSON updated on drop) · Dashboard spec ↔ query mapping layer complete · Anomaly overlay engine — reads GOLD_TREND_BASELINE, renders badges on widgets
Week 13	Dashboard save flow — Draft, Saved, Shared states · Save dialog — name, visibility, tags · Tenant template library — platform templates seeded · Dashboard listing page — grid with previews, filters, search · Share modal — invite by email, set viewer/editor permission
Week 14	Scheduled dashboard — frequency picker, recipient list, PDF snapshot config · Dashboard duplication + copy-to-tenant · Full canvas test — 12-widget dashboard loads in < 3 seconds · Theming — all chart colours respect tenant CSS variables · Widget-by-widget builder end-to-end tested with real data

Phase 3 Exit Criteria

- All 10 widget types working with live data from Snowflake Gold layer
- Dashboard canvas: drag, drop, resize all persist correctly to spec JSON
- Save, share, and template system fully functional
- Anomaly overlay surfacing on correct widgets — verified against baseline
- Tenant branding (colours) applied correctly to all charts

P4 AI Layer — Claude API, Prompt-to-Dashboard

Weeks 15 – 17

The AI layer is where MarketIQ becomes a product rather than a tool. The most important engineering work here is reliable spec generation — the prompts must produce consistent, well-structured JSON across varied user inputs.

Week	Tasks & Deliverables
Week 15	AI Service — GraphQL subgraph + Cloud Run deployment · Claude API integration (@anthropic-ai/sdk) · Intent parsing prompt — extract metrics, dimensions, time range, filters · Spec generation prompt — produce structured JSON dashboard spec from intent · Prompt engineering iteration — test against 20+ varied inputs, tune for consistency
Week 16	Spec → query mapping — AI spec widget configs map to Query Service calls · Parallel query execution — all widget queries run simultaneously · Streaming AI response — spec chunks stream to frontend as

	generated · Dashboard renders progressively as each widget query completes · Refinement loop — subsequent prompts update only affected widgets
Week 17	Full prompt-to-dashboard end-to-end tested with 5 diverse prompts · AI-generated dashboard saves correctly — same spec format as manual builder · Recommendation engine enhanced — Claude generates natural language summaries of anomalies · Prompt bar always visible on canvas — mode-less, available at any time · Performance: full 8-widget dashboard from prompt in < 15 seconds

Phase 4 Exit Criteria

- Prompt-to-dashboard end-to-end working — prompt produces rendered dashboard with live data
- AI-generated dashboards saved and reloaded correctly
- Refinement loop working — iterative prompts modify specific widgets
- Spec generation consistent across 95%+ of test inputs

P5 MVP Hardening & Beta Launch — Weeks 18 – 20

This phase turns a functional product into a launchable one. Beta customers onboarded here provide the first real-world feedback before the full v1 feature build.

Week	Tasks & Deliverables
Week 18	Full onboarding flow — 5-step journey implemented (profile, connectors, sync, first dashboard, invite) · Onboarding progress tracker · Starter dashboard auto-generation for new tenants · Email notifications — data ready, anomaly alerts, scheduled snapshots (Resend + React Email) · Billing integration — Stripe subscription management
Week 19	End-to-end testing across Agency and Enterprise tenant types · Security review — RLS verified, RBAC penetration tested, token rotation checked · Performance baseline — load test with 10 concurrent tenants · Bug fixes from internal testing · Monitoring + alerting setup (Cloud Monitoring, Sentry alert thresholds)
Week 20	Beta customer 1 onboarded — full white-glove setup · Beta customer 2 onboarded · Feedback collection — in-app prompt + direct interviews · Critical feedback → hot fixes · First revenue target reached · MVP retrospective — prioritise Phase 6 backlog

Phase 5 Exit Criteria — MVP Definition

- Two or more paying beta customers live on the platform
- End-to-end onboarding tested with a real agency customer in < 5 days
- All critical and high-severity bugs resolved
- Monitoring and alerting operational — no blind spots in production
- First monthly recurring revenue target met

P6 Full v1 Feature Buildout — Weeks 21 – 26

Phase 6 completes the full v1 vision — remaining connectors, white-label subdomain routing, enterprise RBAC hierarchy, i18n framework, and scheduled report delivery. Feature priority should be adjusted based on beta customer feedback from Phase 5.

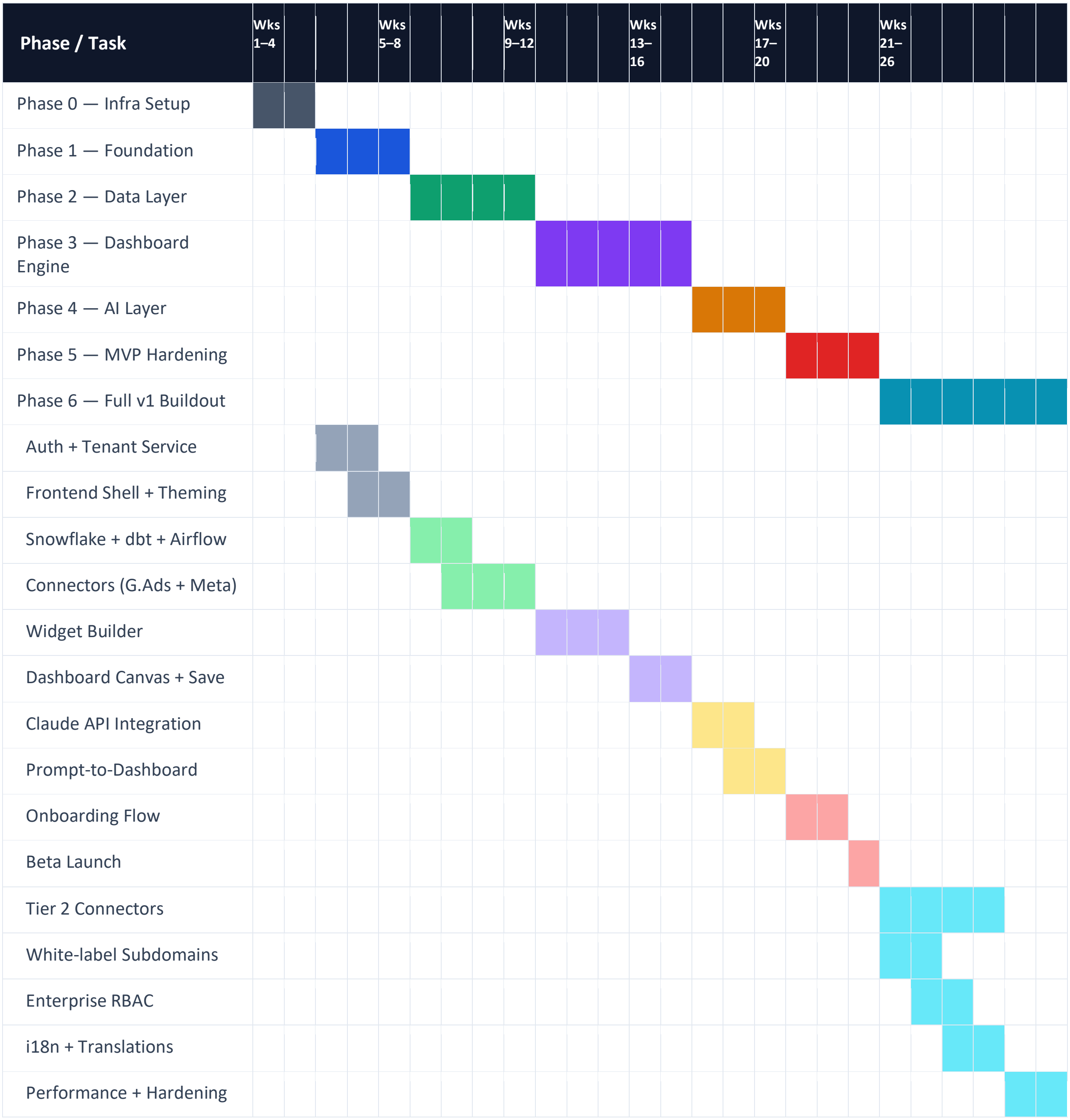
Week	Tasks & Deliverables
Week 21	LinkedIn Ads connector · TikTok Ads connector · White-label subdomain routing — wildcard DNS + SSL automation via Terraform · Branded login page per tenant · Custom domain verification flow
Week 22	Enterprise RBAC hierarchy — Brand/Region/BU topology · Manager role scoping · Enterprise onboarding flow (org structure setup) · Multi-brand/region dashboard roll-ups · Hierarchy permission enforcement
Week 23	Klaviyo connector · Shopify connector · Google Search Console connector · Connector health dashboard — sync status, error log, retry controls · i18n Layer 1 — react-i18next fully wired, all UI strings externalised to locale files
Week 24	KPI translation dictionary — 100 standard marketing KPIs in 3 languages · Dashboard header translation via Azure Translator · PDF export enhancements — branded headers, tenant logo, scheduled delivery · Report scheduling UI — frequency, recipients, format picker
Week 25	Platform template library expansion — 10 templates across 4 verticals · Template customisation flow · Dashboard comparison mode — period-over-period side by side · Full RBAC matrix tested end-to-end across all role combinations
Week 26	Performance optimisation — query caching tuning, frontend bundle analysis · Full security audit · Documentation — API docs, connector setup guides, admin guide · Production hardening — auto-scaling policies, Snowflake cost controls · Full v1 launch

Phase 6 Exit Criteria — Full v1 Definition

- All 5 Tier 1 connectors plus at least 3 Tier 2 connectors live
- White-label custom subdomains working for at least 2 tenants
- Enterprise and Agency tenant types fully functional with correct RBAC
- i18n Layer 1 complete — UI fully externalised and translatable
- Scheduled PDF reports delivered reliably on configurable cadence
- 10+ paying customers on the platform

9.7 Timeline Overview — Gantt Chart

Each row represents a phase or major workstream. Coloured blocks show active weeks.



9.8 Team Size Impact on Timeline

Team Composition	Realistic Timeline	Notes

Solo founder	26–32 weeks to full v1	Context switching + sole decision-making adds 20–25% overhead. MVP achievable at 22 weeks.
Founder + 1 senior full-stack	18–20 weeks to full v1	Frontend and backend can run in parallel from Phase 3 onwards. Recommended minimum.
Founder + 1 frontend + 1 backend	14–16 weeks to full v1	Full parallelism from Phase 2. Fastest route to MVP.
Above + dedicated data engineer	12–14 weeks to full v1	Data layer (Phase 2) runs fully in parallel. Cuts 2–3 weeks off critical path.
Above + QA / DevOps engineer	10–12 weeks to full v1	Testing, CI/CD, and hardening stop blocking feature work.

9.9 MVP vs Full v1 Scope

Feature Area	MVP Scope (Weeks 1–20)	Full v1 Scope (Weeks 21–26)
Connectors	Google Ads + Meta only	GA4, HubSpot, Salesforce, LinkedIn, TikTok, Klaviyo, Shopify, GSC
Dashboard creation	Manual widget builder + AI prompt	Same + template library expansion
Widget types	All 10 types	Same
Tenant types	Agency + Enterprise (basic)	Full Enterprise hierarchy (brand/region/BU)
White-label	Custom logo + colours (no custom domain)	Custom subdomain routing + branded login page
RBAC	Admin + Analyst + Viewer	Full Manager role + Enterprise hierarchy scoping
Language	English only	i18n Layer 1 complete — UI strings externalised
Reports	Manual PDF download	Scheduled PDF delivery to recipient list
Billing	Stripe subscription (basic plans)	Usage-based metering + plan management UI

9.10 Critical Path

These items are blockers. Delay in any one of them delays every subsequent phase:

Critical Path Item	Dependency & Risk
GCP + Snowflake setup (Week 1–2)	Everything builds on this. GCP provisioning delays — budget 2 extra days for account setup.
Snowflake dbt Gold layer complete (Week 9)	AI layer (Phase 4) and Dashboard engine (Phase 3) both depend on Gold layer being queryable.

Connector normalisation layer (Week 7–9)	Every dashboard widget depends on normalised data. Schema decisions made here are hard to change later.
Dashboard spec JSON schema (Week 10)	Both manual builder and AI prompt must produce the same spec format. Define it precisely before building either.
Claude API prompt tuning (Week 15–16)	Reliable spec generation requires iteration. Reserve full 2 weeks — do not compress.

Claude Code usage guidance for this project:

Best for: Prisma schema generation · GraphQL resolver scaffolding · shadcn/ui page layouts
dbt model generation · Terraform module boilerplate · React widget components
Test suite generation · TypeScript type wiring across packages

Not a substitute for: Architecture decisions · Integration debugging · Prompt engineering
Cloud console configuration · Third-party API quirk resolution

Estimated time saving: 35–40% on coding time. Does not reduce architecture or debug time.

10. Infrastructure Cost Estimates

All costs in USD. GCP was selected over Azure specifically to reduce infrastructure cost for a white-label SaaS where cloud brand is invisible to customers. Estimated savings vs equivalent Azure stack: 35–40% at production scale.

10.1 Development Environment

GCP Service	Configuration	Monthly Cost (USD)
Snowflake	Developer edition — paused when not in use (16hrs/day saving)	$40–60
Cloud Run	All services — scale to zero, dev traffic only	$30–40
Cloud Functions	Minimal invocations during development	$5
Cloud SQL PostgreSQL	db-f1-micro — 1 vCPU, 0.6 GB RAM	$10
Memorystore Redis	Basic tier, 1 GB	$35
Cloud Pub/Sub	Low message volume	$5
Firebase Auth	Under 50,000 MAU	Free
Cloud Storage	Data lake + exports, minimal volume	$5
Cloud CDN + Load Balancer	Standard tier	$20
Cloud Monitoring + Logging	Basic retention	$10
Artifact Registry	Basic — image storage	$5
Anthropic Claude API	Development + testing usage	$20–50
Resend (email)	Developer plan	$0–20
Total Development		$185–265/month

> Cost tip: Pause Snowflake capacity during non-working hours and weekends.
> Active ~72 hours/week instead of 168 hours = ~57% Snowflake cost reduction in development.

10.2 Production Environment (50 Tenants)

GCP Service	Configuration	Monthly Cost (USD)
Snowflake	Standard edition — multi-cluster warehouse, S size	$400–600
Cloud Run	Production load — minimum 1 replica on gateway only	$200–350

Cloud Functions	Production invocations — syncs, exports, notifications	$30–50
Cloud SQL PostgreSQL	db-n1-standard-2 — 2 vCPU, 7.5 GB RAM	$150
Memorystore Redis	Standard tier, 5 GB — with replica	$120
Cloud Pub/Sub	Production message volume	$20
Firebase Auth	Per-MAU billing above 50,000 MAU	$30–50
Cloud Storage	Data lake + exports, moderate volume	$40
Cloud CDN + Load Balancer + Cloud Armor	Standard + WAF protection	$80
Cloud Monitoring + Logging	Standard retention — 30 days logs	$60
Anthropic Claude API	Production usage — dashboard generation at scale	$200–500
Resend (email)	Production plan — transactional + scheduled reports	$30
Miscellaneous	Secret Manager, Artifact Registry, networking	$30
Total Production (50 Tenants)		$1,390–2,080/month

10.3 Scaling Guide

Tenant Count	Snowflake Config	Est. Monthly Cost	Recommended Plan Mix
1–10 tenants	Standard XS	$600–800/month	Starter + Growth plans
10–50 tenants	Standard S	$1,400–2,100/month	Growth + Scale plans
50–200 tenants	Standard M	$3,000–4,500/month	Scale + Enterprise plans
200+ tenants	Standard L+	$6,000+/month	Enterprise dominant

10.4 Unit Economics

Scenario	Monthly Revenue	Gross Margin
10 tenants avg $799/mo	$7,990 MRR vs ~$800 infra	~90%
50 tenants avg $900/mo	$45,000 MRR vs ~$2,000 infra	~95.6%
200 tenants avg $1,000/mo	$200,000 MRR vs ~$4,500 infra	~97.8%

11. Security & Compliance

11.1 Security Architecture

Layer	Implementation
Authentication	Firebase Auth — short-lived JWTs, refresh token rotation, secure cookie storage
Authorisation	Role + scope claims in JWT, validated in middleware before every resolver executes
Data isolation	Row-level security in Snowflake — tenant_id injected on every query, enforced at DB layer
Credentials storage	Connector OAuth tokens stored in GCP Secret Manager — encrypted at rest, accessed only by Connector Service
Transport	TLS 1.3 on all endpoints — HSTS enforced, no HTTP fallback
API security	Rate limiting per tenant per endpoint — implemented in Redis via BullMQ
DDoS protection	Google Cloud Armor on all public-facing endpoints — rules updated automatically
Audit logging	All data access events, config changes, and user actions logged to Cloud Logging with 90-day retention

11.2 Compliance Posture

Standard	Status	Implementation
GDPR	In scope — Year 1	Data residency controls · Right to erasure pipeline · DPA templates for customers · Consent logging
SOC 2 Type II	Target — Year 1	Audit logging in place · Access controls documented · Encryption at rest and in transit · Vendor management
Data encryption	Live at launch	AES-256 at rest (GCP default) · TLS 1.3 in transit · Field-level encryption for connector credentials
Connector access	Live at launch	Read-only OAuth scopes only — MarketIQ never writes to or modifies customer ad accounts
Tenant isolation	Live at launch	Workspace-per-tenant in Snowflake · No shared schemas · Cross-tenant queries architecturally blocked

12. Pricing Model

12.1 Plan Structure

Plan	Target	Monthly Price	Key Inclusions
Starter	Small agencies — up to 5 clients	$299/month	5 clients · 3 connectors · 2 users · Basic white-label (logo + colours)
Growth	Mid agencies + SME enterprises	$799/month	25 clients · 8 connectors · 10 users · Full white-label · AI dashboards · Scheduled reports
Scale	Large agencies + enterprise	$1,999/month	Unlimited clients · All connectors · Unlimited users · Custom domain · SSO · Priority support
Enterprise	Global agencies + large enterprise	Custom	Dedicated infra · Custom SLA · Custom connectors · Dedicated onboarding · White-glove support

12.2 White-Label Resale Model

Agencies purchase a Growth or Scale plan, then resell access to their clients at their own pricing. The agency sets the client's price — MarketIQ has no direct relationship with the end client. This mirrors how agencies currently resell analytics and reporting tools.

Example agency economics:

Agency pays MarketIQ: $799/month (Growth plan — 25 clients)

Agency charges each client: $300/month (client-branded reporting tool)

10 clients × $300 = $3,000/month client revenue

Net margin to agency from platform alone: $3,000 - $799 = $2,201/month

13. Success Metrics

13.1 Product KPIs

Metric	Target	Measurement
Onboarding completion rate	Greater than 70%	Sign-up to first saved dashboard / total sign-ups
Time to first dashboard	5 days maximum	Timestamp: sign-up → first dashboard saved
Dashboard creation — AI prompt	Under 30 seconds	Time from prompt submission to full render
Dashboard creation — manual builder	Under 10 minutes	Time from canvas open to first save
Gold layer query response time	Under 2 seconds p95	Measured via Cloud Monitoring on Query Service
API response time	Under 500ms p95	Measured across all GraphQL resolvers
Data sync reliability	99.5% uptime	Connector sync success rate per week
AI spec generation success rate	Greater than 95%	Valid renderable spec produced / total prompts

13.2 Business KPIs

Metric	Target	Timeframe
Monthly Recurring Revenue (MRR)	$25,000+	Month 6 from launch
Paying tenants	15+	Month 6 from launch
Net Revenue Retention (NRR)	Greater than 110%	Rolling 12-month
Gross Margin	Greater than 85%	Once 10+ tenants live
Payback period	Under 3 months CAC	Rolling
Time to second dashboard	Under 7 days	From first dashboard creation

14. Open Decisions & Risks

14.1 Open Decisions

Decision	Options	Recommendation
Managed Airflow	Cloud Composer vs Astronomer.io	Astronomer.io — lower cost, simpler ops, same Apache Airflow API
PDF export	Self-hosted Puppeteer vs Browserless.io	Browserless.io — managed, saves 3–4 days of infra setup
Email service	GCP Communication Services vs Resend	Resend — developer-friendly API, React Email templates, easier v1
Billing integration	Stripe vs Paddle	Stripe — simpler, most widely supported, best documentation
Snowflake vs BigQuery	Snowflake (current choice) vs BigQuery (cheaper)	Snowflake — better multi-tenant isolation, richer SQL, worth the cost premium

14.2 Risk Register

Risk	Likelihood	Impact	Mitigation
Connector API breaking changes disrupting data pipelines	Medium	High	Airbyte community connectors maintained frequently · Monitor deprecation notices · Abstract behind normalisation layer
Snowflake costs exceeding estimates at scale	Low	Medium	Consumption-based — implement per-tenant query cost controls and monthly budget alerts
Claude API latency in dashboard generation (>15 seconds)	Low	Medium	Cache generated specs for identical prompts · Stream AI response progressively · Set user expectation with loading animation
Multi-tenant data leak via misconfigured RLS	Very Low	Critical	RLS enforced at DB layer not app layer · Penetration test before launch · Automated RLS regression tests
Timeline slippage on data layer (most complex phase)	Medium	High	Phase 2 is the critical path — start Snowflake and dbt in parallel from Week 6 · Hire data engineer if slipping
Solo development pace slower than estimated	Medium	Medium	MVP scope reduction available (cut Tier 2 connectors, scheduled reports) · First revenue target at Week 20 not Week 26

End of Document

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